

# Camden Conrad

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U.S. Citizen · Security Clearance: None (Eligible)

Rust systems programmer specializing in astrodynamics and deterministic flight software. Author of *Orbital*, an interplanetary trajectory optimizer that rediscovers the Mars 2020 launch window to within a day of departure, and *Besom*, a deterministic simulation harness for NASA's core Flight System (cFS). Shipped commercial software (*Axis Football 2026*, 10K+ active players) and daily-drives a self-built Linux stack. Focused on orbital mechanics, GNC, determinism, and zero-allocation real-time design — software written as if a bit flipped somewhere.

## TECHNICAL SKILLS

**Aerospace & GNC:** Orbital Mechanics · Trajectory Optimization · Lambert / Porkchop · Gravity-Assist Tours · B-Plane Targeting · N-Body & Post-Newtonian (1PN) Dynamics · DP5(4) Integration · JPL DE440 / SPICE (ANISE) · Deterministic Simulation · NASA cFS/OSAL

**Languages:** Rust · C · C++ · C# · Python · SQL · TypeScript · WGSL · HLSL

**Performance & Systems:** Zero-Allocation Hot Paths · Manual Memory Management · SIMD · Multithreading · Thread Safety · Data-Oriented Design · Perf Profiling · Wayland/Smithay · sched\_ext · Instrumentation

**Graphics & Tooling:** wgpu · Custom Renderers · WGSL/HLSL Shader Pipelines · GPU Compute (ILGPU) · Custom LSP/Debugger Clients · Static Analysis · Power of Ten

## PROFESSIONAL EXPERIENCE

**Junior Software Engineer** · Farrell Holdings Inc · Remote

May 2026 – Present

- Full-stack development of internal tooling across the web stack (JavaScript/TypeScript, Python, Rust) — owning features end-to-end from UI through backend services and data layers.

**Contract Systems Programmer** · Axis Games · Remote · Axis Football 2026

April 2025 – May 2026

- Shipped *Axis Football 2026* on PC & console (10,000+ active players) as a credited Contract Systems Programmer.
- Rebuilt the replay system around zero-allocation hot paths — 15 s of full 22-player state in 25 MB with variable-speed playback; instrumentation closed out the timing-related crash class QA was hitting.
- Re-architected the crowd-editor pipeline for a 7.8× speedup, cutting designer iteration from days to minutes while supporting 10× larger crowds.
- Automated the live-statistics content pipeline — parses real-time player/season data into structured narrative commentary at scale.

## FEATURED PROJECTS

**Orbital — Interplanetary Trajectory Optimizer & Solar-System Simulator** — Rust · wgpu · ANISE/DE440 · Apache-2.0 ·

[github.com/camdenconrad/Orbital](https://github.com/camdenconrad/Orbital)

- Designs real interplanetary transfers under full n-body, post-Newtonian (1PN) dynamics — Lambert/porkchop screening, a Lambert-seeded beam search, gravity-assist tours, low-thrust (SEP) profiles, and a two-phase corrector with B-plane targeting at arrival.
- Validated against published mission data — rediscovers the Mars 2020 launch to within one day of departure and ~1% of launch energy, and finds the 2026/27 type-II Mars window consistent with the JPL Trajectory Browser.
- Propagates on JPL DE440 ephemerides (via ANISE) with a DP5(4) integrator and GPU porkchop screening — a single native Rust app (eframe/egui + wgpu).

**Besom — Deterministic Simulation Harness for NASA cFS** — Rust · NASA cFS/OSAL · Apache-2.0 · [github.com/camdenconrad/besom](https://github.com/camdenconrad/besom)

- Runs real, unmodified cFE flight software but owns the clock — time advances only per granted tick, so identical scenarios produce byte-identical telemetry, making flight software regression-testable in a way a real-clock harness (e.g. NASA's NOS3) cannot. Doubles as a ground station.

**Cauldron — GPU-Rendered Rust IDE for Flight-Software Discipline** — Rust · wgpu · Static Analysis ·

[github.com/camdenconrad/cauldron](https://github.com/camdenconrad/cauldron)

- Checks NASA/JPL coding rules (Power of Ten, JPL C) as live save-time diagnostics, including whole-project call-graph analysis such as recursion detection that a language server alone cannot express; replicates cFS CI checks locally.

**RuneOS — Custom Linux Environment (Daily Driver)** — Rust · Wayland/Smithay · sched\_ext

- Daily-drives a full self-built stack: a packaged kernel (linux-rune), a Wayland compositor on Smithay, a native Rust shell, a package manager, and ~30 system daemons — with portions held to the JPL coding standard.

**repliKate — High-Velocity Learning System** — C# · Custom ML · ILGPU · GPU Acceleration

- Lightweight continuous-learning engine — adapts without pretraining at ~100 MB RAM. Thread-safe online-learning pipeline applied to Chess AI (MCTS), game AI (A\* pathfinding), and ERA5 / NASA POWER weather prediction.

## EDUCATION

**University of North Carolina at Charlotte**

2021 – 2026

Computer Science, AI / Robotics & Game Development concentration — completed the CS core curriculum and senior project.